Determinacy Algebra: Being, Becoming, and Laws That Bite

Executive summary. We model "being" as closure under an idempotent operator α. The size $F = |\mathrm{Fix}(\alpha)|$ of its determinate core satisfies a submultiplicative law under constraint-coupled composition: $F(X \widehat{\otimes}_K Y) \le F(X) \cdot F(Y)$, with **defect **$\Delta$ = |Forbidden determinate pairs|. This defect quantifies how laws bite. The framework lifts cleanly to presheaves for context-dependent semantics and admits temporalization for becoming.

Core Definitions & Theorems

Assumption. We work primarily in the finite setting; cardinal arithmetic extends the results.

Definition 1 (Mode Object). A carrier X with idempotent $\alpha : X \to X$ ($\alpha \circ \alpha = \alpha$). Its **determinate core** is $\mathrm{Fix}(\alpha) = \{x \mid \alpha(x) = x\}$. The **size invariant** $F(X, \alpha) := |\mathrm{Fix}(\alpha)| \in \mathbb{N}$ (or Card).

Definition 2 (Law & Constrained Tensor). For (X, α_X), (Y, α_Y), a **law** $K \subseteq X \times Y$ is closed under $\alpha_X \times \alpha_Y$. The **constrained tensor** is the subobject $K \hookrightarrow X \times Y$ equipped with $\alpha_K := (\alpha_X \times \alpha_Y)\restriction_K$, which remains idempotent because $(\alpha_X \times \alpha_Y)^2 = \alpha_X \times \alpha_Y$ and K is closed under $\alpha_X \times \alpha_Y$.

The Three Key Theorems

Theorem 1 (Fixed-Point Formula).

$$ \mathrm{Fix}\big((X,\alpha_X) \, \widehat{\otimes}_K \, (Y,\alpha_Y)\big) = \big(\mathrm{Fix}(\alpha_X) \times \mathrm{Fix}(\alpha_Y)\big) \cap K. $$

Theorem 2 (Submultiplicativity & Equality Criterion).

$$ F\big((X,\alpha_X) \, \widehat{\otimes}_K \, (Y,\alpha_Y)\big) \le F(X,\alpha_X)\cdot F(Y,\alpha_Y). $$

Equality holds **iff** $\mathrm{Fix}(\alpha_X) \times \mathrm{Fix}(\alpha_Y) \subseteq K$, i.e., when the inclusion $(\mathrm{Fix}(\alpha_X) \times \mathrm{Fix}(\alpha_Y)) \cap K \hookrightarrow \mathrm{Fix}(\alpha_X) \times \mathrm{Fix}(\alpha_Y)$ is an isomorphism.

Theorem 3 (Defect Identity). Define the **multiplicativity defect**

$$ \Delta_K(X,Y) := F(X)F(Y) - F\big((X,\alpha_X)\widehat{\otimes}_K(Y,\alpha_Y)\big). $$

Then:

$$ \Delta_K(X,Y) = |\,(\mathrm{Fix}(\alpha_X)\times\mathrm{Fix}(\alpha_Y)) \setminus K\,|. $$

Thus F is a **lax monoidal** size functor with defect Δ.

Additional Properties

Proposition (Determinacy monotonicity). If $\mathrm{Fix}(\alpha_X) \subseteq \mathrm{Fix}(\alpha'_X)$ and $\mathrm{Fix}(\alpha_Y) \subseteq \mathrm{Fix}(\alpha'_Y)$, then

$$ F\big((X,\alpha_X)\widehat{\otimes}_K(Y,\alpha_Y)\big) \le F\big((X,\alpha'_X)\widehat{\otimes}_K(Y, \alpha'_Y)\big). $$

Proposition (Law monotonicity). If $K \subseteq K'$ then

$$ F\big((X,\alpha_X)\widehat{\otimes}K(Y,\alpha_Y)\big) \le F\big((X,\alpha_X)\widehat{\otimes}(Y,\alpha_Y) \big), \quad \Delta_K(X,Y) \ge \Delta_{K'}(X,Y). $$

Theorem (Defect subadditivity under intersections). For any $K, L \subseteq X \times Y$ closed under $\alpha_X \times \alpha_Y$,

$$ \Delta_{K\cap L}(X,Y) \le \Delta_K(X,Y) + \Delta_L(X,Y). $$

Proof. The forbidden set for $K \cap L$ is the union of the two forbidden sets; apply $|A \cup B| \le |A| + |B|$.

Proposition (Defect under unions). For K, L closed under $\alpha_X \times \alpha_Y$,

$$ \Delta_{K\cup L}(X,Y) \ge \max{\Delta_K(X,Y),\ \Delta_L(X,Y)}. $$

Bonferroni (two laws).

$$ \Delta_{K\cap L}(X,Y) \ge \Delta_K(X,Y)+\Delta_L(X,Y)-\Delta_{K\cup L}(X,Y). $$

Definition (n-ary law and defect). For (X_i, α_i) $(i = 1, \dots, n)$ and $K \subseteq \prod_i X_i$ closed under $\prod_i \alpha_i$,

$$ \widehat{\bigotimes}_{K}\, (X_1,\alpha_1), \dots, (X_n,\alpha_n) := \big(K, (\prod_i \alpha_i)! \restriction_K\big), $$

$$ \Delta_K(X_1,\dots,X_n) := \prod_i F(X_i) - F!\big(\widehat{\bigotimes}_{K} (X_i,\alpha_i)\big) = \big|\, (\prod_i \mathrm{Fix}(\alpha_i)) \setminus K \,\big|. $$

Associativity & the Associator Defect

When laws are only pairwise, associativity is not automatic.

Setup. Let $K \subseteq X \times Y,\ L \subseteq Y \times Z,\ M \subseteq X \times Z$ (all closed under the respective α's). Define two composites:

$$ \begin{aligned} F_{(XY)Z} &:= F\big((X\widehat{\otimes}*K Y) \widehat{\otimes}* Z\big),\
F_{X(YZ)} &:= F\big(X \widehat{\otimes}_{\widetilde K} (Y\widehat{\otimes}_L Z)\big), \end{aligned} $$

where $\widetilde{L}, \widetilde{K}$ are the induced laws on triples obtained by pullback/pushforward along projections.

Definition (Associator defect).

$$ \delta_{\mathrm{assoc}}(X,Y,Z;K,L,M) := |F_{(XY)Z} - F_{X(YZ)}|. $$

Theorem (Associativity via 3-ary extension). $\delta_{\text{assoc}} = 0$ **iff** there exists a $T \subseteq X \times Y \times Z$ closed under $\alpha_X \times \alpha_Y \times \alpha_Z$ such that $\pi_{XY}[T] = K, \quad \pi_{YZ}[T] = L, \quad \pi_{XZ}[T] = M$, and both composites are the same subobject $(T, (\alpha_X \times \alpha_Y \times \alpha_Z)\restriction_T)$ viewed along the two bracketings.

Why it matters: δ_{assoc} flags non-associative constraint interactions—critical for systems engineering.

The Bool Witness Family

Take $\mathbf{Bool}$ with $\alpha = \mathrm{id}$ (everything determinate). Then $F(\mathbf{Bool}) = 2$.

Law K	Description	Forbidden pairs	$F(\mathbf{Bool}\widehat{\otimes}_K\mathbf{Bool})$	Δ
All-allowed ($X \times Y$)	No law	$\varnothing$	4	0
Implication ($b_1 \Rightarrow b_2$)	Entailment	(1,0)	3	1
At-most-one ($\neg(b_1 \wedge b_2)$)	Exclusion	(1,1)	3	1
At-least-one ($b_1 \vee b_2$)	Coverage	(0,0)	3	1
Diagonal ($b_1 = b_2$)	Agreement	(0,1), (1,0)	2	2
XOR ($b_1 \oplus b_2 = 1$)	Incompatibility	(0,0), (1,1)	2	2

Different laws yield different defects, all computable via Theorem 3.

Presheaf Lift: Global Actual Being

Let P be a poset (stages, worlds, contexts). Work in $\mathcal{C} = \mathbf{Set}^{P^{\mathrm{op}}}$.
Notation: Terminal presheaf 1; global sections $\Gamma(Z) = \mathrm{Nat}(1, Z)$.
Equalizers and monos are computed **pointwise**.

A **presheaf mode object** is (X,α) with $\alpha : X \Rightarrow X$ natural and idempotent.
Define $\mathrm{Fix}(\alpha)$ as the equalizer presheaf (pointwise).
Define **actual being** as $F(X,\alpha) := |\Gamma(\mathrm{Fix}(\alpha))|$.

Choice of aggregator. We take $F = |\Gamma(\mathrm{Fix}(\alpha))|$ to measure **coherent** being across contexts. Alternatives (pointwise counts, colimits) capture potential/aggregate being. Results are stable under switching to any aggregator that is a limit-preserving functor out of $\mathbf{Set}^{P^{\mathrm{op}}}$.

Example (two stages). For $P = \{0 \to 1\}$, with restriction $r_{10} : X(1) \to X(0)$,

$$ \Gamma(\mathrm{Fix}(\alpha))\ \cong\ {x_1\in \mathrm{Fix}(\alpha(1))\mid r_{10}(x_1)\in \mathrm{Fix}(\alpha(0))}, $$

so $F \leq \min\{\#\mathrm{Fix}(\alpha(0)), \#\mathrm{Fix}(\alpha(1))\}$.

Lemma (Constant Lift). For constant presheaves and pointwise K, $\Gamma(\mathrm{Fix}(\alpha)) \cong \mathrm{Fix}(\alpha)$ at any stage. Hence the Bool witnesses lift unchanged.

Context P	Interpretation	$\Gamma(\mathrm{Fix}(\alpha))$ meaning
Single point	Static world	Actual core
$(\mathbb{N}, \leq)$	Developmental stages	Stable identity across stages
Kripke frame	Modal worlds	Necessary core (world-invariant)
Intervention poset	Causal contexts	Dispositions surviving all interventions

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Introduce a time-indexed family $(\alpha_t)_{t\in T}$ with $\alpha_s \leq \alpha_t$ pointwise ($s \leq t$).

Lemma (Monotone Cores). $\mathrm{Fix}(\alpha_s) \subseteq \mathrm{Fix}(\alpha_t) \Rightarrow F_s \leq F_t$.

Process law: $\alpha_{t+1} \circ \alpha_t = \alpha_{t+1}$ (lax idempotence across time).
Event: strict increase in F_t.
Becoming-two time: $t^* := \min\{t \mid F_t = 2\}$ (when exists).

Lemma (Event–defect identity). For fixed K,

$$ \Delta_{K,t} = \big|\,(\mathrm{Fix}(\alpha_{X,t})\times\mathrm{Fix}(\alpha_{Y,t})) \setminus K\,\big|. $$

Corollary (Stability). If K is constant in t and α_t stabilizes, then $\Delta_{K,t}$ stabilizes.

Law Taxonomy & Closure Operator

- **α****-preserving:** $(x,y)\in K \Rightarrow (\alpha_X x, \alpha_Y y)\in K$.
- **α****-creating:** $\exists\ \alpha' \ge \alpha$ with K α'-preserving but not α-preserving.
- **α****-violating:** references indeterminate bits (disallowed if α encodes lawful sector).

Law-closure operator.

$$ \mathrm{LawCl}_\alpha(K) := \min\{\alpha' \ge \alpha \mid K \text{ is } \alpha'\text{-preserving}\}. $$

Monotonicity:
$K \subseteq K' \Rightarrow \mathrm{LawCl}_\alpha(K) \le \mathrm{LawCl}_\alpha(K')$.
$\alpha \le \alpha' \Rightarrow \mathrm{LawCl}_\alpha(K) \le \mathrm{LawCl}_{\alpha'}(K)$.

Proposition (Law closure is a nucleus). Ordering idempotents by fixed-set inclusion, LawCl_α is inflationary, monotone, and idempotent; hence a nucleus on this poset. The image is a reflective subposet of determinacy levels where K is admissible.

Remark (Karoubi split). Every idempotent α on $\mathbf{Set}$ splits; (X,α) is equivalent to the inclusion $i : \mathrm{Fix}(\alpha) \hookrightarrow X$. In this normal form, Theorem 1 becomes "intersect with K then count," and all other results reduce to elementary set arithmetic.

Unit & Isomorphism Invariance

Unit mode object. Let $\mathbb{I}$ be a one-element set with $\alpha_{\mathbb{I}} = \mathrm{id}$. For any law $K \subseteq X \times \mathbb{I}$ closed under $\alpha_X \times \alpha_{\mathbb{I}}$, we have $K \simeq X$, $\alpha_K \simeq \alpha_X$, hence

$$ F\big((X,\alpha_X)\widehat{\otimes}K(\mathbb I,\alpha)\big)=F(X,\alpha_X). $$

Taking $K = X \times \mathbb{I}$ realizes $\mathbb{I}$ as the tensor unit.

Isomorphism invariance. If $f : (X,\alpha_X) \to (X',\alpha_{X'})$ is an isomorphism commuting with α, then F and Δ_K are preserved under transport along $f \times g$.

Computational Recipe (finite case)

Algorithm (finite Δ). Precompute $A := \mathrm{Fix}(\alpha_X)$, $B := \mathrm{Fix}(\alpha_Y)$. Then
$F(X\widehat{\otimes}_K Y) = \#\{(a,b) \in A \times B :\ (a,b) \in K\}$, and $\Delta = \#(A \times B) - F$.
Cost: $O(|A||B|)$ membership checks in K. With bitsets/sorted lists, stream with $O(|A| + |B|)$ memory.

Lean-Flavored Code Snippets

Set-level skeleton (finite case)

```
-- Finite mode objects with explicit cardinality
structure ModeObj (X : Type) [Fintype X] where
  α : X → X
  idem : ∀ x, α (α x) = α x

namespace ModeObj
variable {X Y : Type} [Fintype X] [Fintype Y] (M : ModeObj X) (N : ModeObj Y)

-- Fixed points as a Finset (helper predicate omitted for brevity)
def Fix : Finset X := {x | M.α x = x}

def F : ℕ := (Fix M).card

structure Law where
  K : Finset (X × Y)
  closed : ∀ p ∈ K, (M.α p.1, N.α p.2) ∈ K

def tensor (L : Law M N) : ModeObj {p // p ∈ L.K} where
  α := fun ⟨p, h⟩ => ⟨(M.α p.1, N.α p.2), L.closed p h⟩
  idem := fun ⟨p, h⟩ => by
    ext <;> simp [M.idem, N.idem]

-- Theorem 1: fixed-point formula (sketch)
theorem fix_tensor (L : Law M N) :
    (tensor L).Fix = (M.Fix ×ˢ N.Fix).filter (· ∈ L.K) := by
  -- unpack definitions; idempotence + closure suffice
  admit

-- Theorem 2: submultiplicativity (sketch)
theorem F_le_mul (L : Law M N) : F (tensor L) ≤ F M * F N := by
  -- compare cardinalities via inclusion into product core
  admit

-- Theorem 3: defect identity (sketch)
def defect (L : Law M N) : ℕ := F M * F N - F (tensor L)

theorem defect_eq (L : Law M N) :
    defect L = ((M.Fix ×ˢ N.Fix) \ L.K).card := by
  -- inclusion–exclusion on filtered product
  admit
```

Bool witness (diagonal law)

```
def BoolMode : ModeObj Bool := ⟨id, λ _ => rfl⟩

def diagLaw : Law BoolMode BoolMode := {
  K := {(true,true), (false,false)},
  closed := by simp
}

#eval BoolMode.F                -- 2
#eval (tensor diagLaw).F        -- 2 (< 4)
#eval defect diagLaw            -- 2
```

(Swap $\mathbb{N}$ for cardinals to remove finiteness; port to presheaves by working in functor categories and using pointwise equalizers.)

Relevant External References

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FAQ (quick checks)

- **Infinite case?** All statements hold in $\mathbf{Card}$; submultiplicativity uses cardinal multiplication; Theorem 3 uses set-difference cardinality.
- **Does F functorialize?** Yes: on morphisms commuting with α, F is monotone on monos; equality holds exactly when the mono is an iso on fixed sets.
- **Why 'tensor'?** It is a law-parametrized product; formally a **lax monoidal** structure where the unit is the mode object with singleton core.

One-sentence value proposition

Determinacy algebra turns “laws reduce independence” into a **lax-monoidal arithmetic with a measurable defect ****Δ, unifying static, modal, and developmental readings and exposing non-associative law interactions as quantifiable risk.